

STA 131A: Introduction to Probability Theory

Lecture 19: Derived Distributions (cont'd) + Review for Midterm 2

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Announcement

Midterm 2 on Fri, May 15 (10:00 am–10:50 am in class)

- **Arrive early:** The exam starts at 10:00 am and ends at 10:50 am sharp
- **One hand-written cheat sheet:** Letter-size (8.5"×11"), double-sided, brief formulas/notes
- **Calculator:** A simple (non-graphing) scientific calculator is allowed
- **No other materials** beyond the single cheat sheet (no textbooks, etc.) is allowed
- **SDC accommodations:** Confirm scheduling with AES online ASAP

Discussion section tomorrow (Thu, May 14)

- The TA will go over two practice midterms

Office hour today (Wed, May 13) 2:30–3:30 PM

Agenda

Last time: Derived distributions

- Functions of one random variable: $Z = g(X)$
 - Generic method: support $\rightarrow$ CDF $\rightarrow$ PDF
 - Monotone transformation formula
 - Linear transformations as a special case
- Functions of two random variables: $Z = g(X, Y)$
 - CDF by integrating over a region

Today: Derived distributions, continued + Midterm 2 review

- Finish functions of two random variables
 - Recap CDF-first method for $Z = h(X, Y)$
 - Convolution for sums of independent random variables
- Brief review for Midterm 2
 - Continuous PDFs/CDFs, normal variables, joint PDFs
 - Conditioning, Bayes, conditional expectation
 - Derived distributions

Recap: Derived distributions

Generic, CDF-first workflow for $Z = g(X)$

1. Identify the support: possible values of $Z = g(X)$.

2. Compute the CDF:

$$F_Z(z) = P(Z \leq z) = P(g(X) \leq z).$$

3. Differentiate on the support:

$$f_Z(z) = \frac{d}{dz} F_Z(z).$$

4. Set $f_Z(z) = 0$ outside the support.

PDF formula for a strictly monotone transformation

Suppose X is continuous and $Z = g(X)$, where g is strictly monotone and differentiable on the support of X , with inverse

$$x = h(z) = g^{-1}(z).$$

Then, for z in the support of Z ,

$$f_Z(z) = f_X(h(z)) \cdot |h'(z)|.$$

Recap: Function of two random variables

Let (X, Y) be jointly continuous, and let

$$Z = g(X, Y).$$

To find the distribution of Z , a general strategy is again to identify the CDF first:

$$F_Z(z) = P(Z \leq z) = P(g(X, Y) \leq z) = \iint_{\{(x,y): g(x,y) \leq z\}} f_{X,Y}(x, y) \, dx \, dy.$$

Then, when differentiable,

$$f_Z(z) = F'_Z(z).$$

Main challenge: Identifying the integration region correctly.

Practical rules: Draw the support of (X, Y) , then draw the boundary $g(x, y) = z$. Split the integral as needed whenever the relevant boundary changes.

- Recall the examples: With (X, Y) uniformly distributed on $[0, 1]^2$, $Z_1 = XY$ or $Z_2 = \frac{Y}{X}$.

Sum of independent random variables: Convolution

Definition (Convolution)

Let (X, Y) be jointly continuous and let $Z = X + Y$.

If X and Y are independent, the density of $Z = X + Y$ is obtained by the **convolution** operation:

$$f_Z(z) = (f_X * f_Y)(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx.$$

- **Support rule:** The integral only includes x 's for which both
$$x \in \text{supp}(X) \quad \text{and} \quad z - x \in \text{supp}(Y).$$
- **Interpretation:** To get $X + Y = z$, sum over all possible ways $x + (z - x) = z$.
- **Proof idea:** Given $X = x$, we have $Z = x + Y$, so

$$f_{Z|X}(z | x) = f_Y(z - x).$$

Thus $f_{X,Z}(x, z) = f_X(x)f_Y(z - x)$, and integrating out x gives the convolution formula.

- The convolution of PMFs is defined similarly: $p_Z(z) = (p_X * p_Y)(z) = \sum_x p_X(x) p_Y(z - x)$.

Example: Sum of two independent uniforms

Example

Let $X, Y \stackrel{\text{independent}}{\sim} \text{Uniform}(0, 1)$, and $Z = X + Y$.

Question: Find the PDF of Z .

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx.$$

Since $f_X(x) = 1$ for $0 < x < 1$ and $f_Y(z - x) = 1$ when $0 < z - x < 1$, the integrand $f_X(x)f_Y(z - x)$ is nonzero only when

$$0 < x < 1 \quad \text{and} \quad z - 1 < x < z.$$

Thus x must lie in the interval

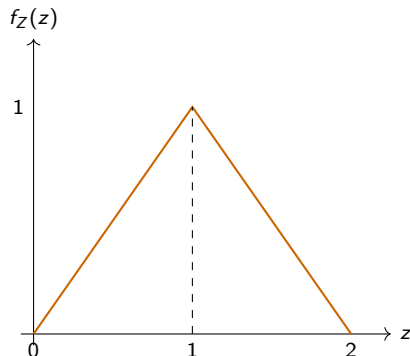
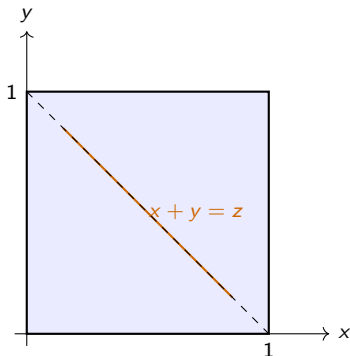
$$\max(0, z - 1) < x < \min(1, z).$$

The density of Z at z is equal to the length of this interval (whenever it is positive):

$$f_Z(z) = \begin{cases} z, & 0 < z < 1, \\ 2 - z, & 1 \leq z < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Geometric view of $X + Y$

For $X, Y \sim \text{Uniform}(0, 1)$, the joint density is uniform on the unit square.



For a thin band $z < X + Y < z + \Delta z$, the probability is the area of the band. As $\Delta z \rightarrow 0$, this area is proportional to the length of the slice $x + y = z$ inside the unit square.

This length increases for $0 < z < 1$ and decreases for $1 < z < 2$, producing a triangular density.

Example: Sum of independent exponentials

Example

Let X, Y be independent $\text{Exponential}(\lambda)$ random variables, and let

$$Z = X + Y.$$

Question: Find the PDF of Z .

Since $X, Y > 0$, we have $Z > 0$. Using convolution, for $z > 0$,

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx \\ &= \lambda^2 e^{-\lambda z} \int_0^z dx = \lambda^2 z e^{-\lambda z}. \end{aligned}$$

Thus,

$$f_Z(z) = \begin{cases} \lambda^2 z e^{-\lambda z}, & z > 0, \\ 0, & z \leq 0. \end{cases}$$

Remark: This distribution is often called an Erlang/Gamma distribution with shape 2.

Pop-up quiz: Convolution

Let X, Y be independent $\text{Uniform}(0, 1)$, and let

$$Z = X - Y.$$

Which describes the density of Z correctly?

A) $f_Z(z) = \frac{1}{2}, \quad -1 < z < 1$

C) $f_Z(z) = 1, \quad -1 < z < 1$

B) $f_Z(z) = \begin{cases} z + 1, & -1 < z < 0, \\ -z + 1, & 0 < z < 1, \\ 0, & \text{otherwise} \end{cases}$

D) $f_Z(z) = z, \quad -1 < z < 1$

Answer: B. Since $Z = X - Y$, write $Y = X - z$. The convolution-type formula is

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(x - z) dx.$$

The integrand is nonzero when $0 < x < 1$ and $0 < x - z < 1$, or equivalently,

$$\max(0, z) < x < \min(1, 1 + z).$$

Midterm 2 review: What to know cold

Core skills:

- Work with PDFs and CDFs:

$$P(a < X \leq b) = F_X(b) - F_X(a) = \int_a^b f_X(x) dx.$$

- Standardize normal random variables:

$$Z = \frac{X - \mu}{\sigma}.$$

- Compute marginals and conditional densities from joint PDFs.
- Use conditional expectation and the law of total expectation.
- Derive distributions using CDF-first approach, monotone PDF formula, or convolution.

Main exam advice: Identify the support and the relevant region before integrating.

Review: Continuous random variables

Continuous random variable and PDF:

$$P(X \in A) = \int_A f_X(x) dx, \quad f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1.$$

CDF:

$$F_X(x) = P(X \leq x), \quad P(a < X \leq b) = F_X(b) - F_X(a).$$

If X is continuous and F_X is differentiable, then

$$F_X(x) = \int_{-\infty}^x f_X(t) dt, \quad f_X(x) = F'_X(x).$$

Warning: A density value is not a probability itself; probabilities come from integration.

Review: Normal random variables

Normal random variable:

$$X \sim N(\mu, \sigma^2)$$

indicates X has mean μ and variance σ^2 (= standard deviation σ).

Standardization:

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

Thus, using the standard normal CDF Φ ,

$$P(a < X \leq b) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right).$$

- Note that $\Phi(z) = 1 - \Phi(-z)$ due to symmetry

Common mistake: in $N(\mu, \sigma^2)$, σ is the standard deviation and σ^2 is the variance.

Review: Joint PDFs and marginalization

Joint PDF:

$$P((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy.$$

Marginals:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx.$$

Expectation:

$$\mathbb{E}[g(X, Y)] = \iint g(x, y) f_{X,Y}(x, y) dx dy.$$

Warning: Integration limits are identified from the geometry of the support.

Review: Conditioning and independence

Conditional PDF:

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}, \quad f_Y(y) > 0.$$

Conditional expectation:

$$\mathbb{E}[X | Y = y] = \int x f_{X|Y}(x | y) dx.$$

Total expectation:

$$\mathbb{E}[X] = \int \mathbb{E}[X | Y = y] f_Y(y) dy.$$

Independence:

$$f_{X,Y}(x, y) = f_X(x)f_Y(y), \quad \forall (x, y).$$

Review: Bayes' rule

Continuous hidden state X :

$$f_{X|Y}(x | y) = \frac{f_X(x)f_{Y|X}(y | x)}{\int f_X(t)f_{Y|X}(y | t) dt}.$$

Discrete hidden state X , continuous observation Y :

$$P(H = h | Y = y) = \frac{P(H = h)f_{Y|H}(y | h)}{\sum_{h'} P(H = h')f_{Y|H}(y | h')}.$$

Implication:

posterior $\propto$ prior $\times$ likelihood.

Review: Derived distributions

Generic CDF-first method:

$$F_Z(z) = P(g(X) \leq z), \quad f_Z(z) = F'_Z(z).$$

Strictly monotone formula: If $Z = g(X)$, g is differentiable and strictly monotone, and $x = h(z) = g^{-1}(z)$, then, for z in the support of Z ,

$$f_Z(z) = f_X(h(z)) |h'(z)|.$$

Function of two variables:

$$F_Z(z) = P(h(X, Y) \leq z) = \iint_{\{(x,y): h(x,y) \leq z\}} f_{X,Y}(x, y) \, dx \, dy.$$

Convolution: If $Z = X + Y$ and X, Y are independent,

$$f_Z(z) = \int f_X(x) f_Y(z - x) \, dx.$$

Midterm 2 review: Tips and common mistakes

- A density value is not a probability; probabilities come from integration.
- For continuous variables, endpoints usually do not matter.
- Sometimes, exploiting symmetry can simplify calculation.
- In $N(\mu, \sigma^2)$, σ^2 is the variance and σ is the standard deviation.
- For joint densities, integration limits come from the geometry of the support.
- Independence requires factorization:

$$f_{X,Y}(x, y) = f_X(x)f_Y(y), \quad \text{for all } (x, y).$$

- For transformations, the support of the transformed random variable must be stated as part of the answer.

Wrap-up: How to prepare

Use the practice midterms for review

- First attempt them without notes and under time pressure.
- Mark which steps failed: identifying the tool, setting up the model, or doing the calculation.
- Then review and prepare your cheat sheet around those weak points.

When preparing your cheat sheet

- Consider including definitions and assumptions, not only formulas.
- Ensure you are fluent with the tools you have written.

During the exam

- Succinctly write down the relevant event or random variable before using a formula.
- Keep notation clear: distinguish events, random variables, PDFs/CDFs, and expectations.
- If stuck, identify the support, draw the region, or condition on a useful variable/event.

Advice: Be fluent in translating words to probabilistic objects and choosing the right tool.